

	NOIDA INSTITUTE OF ENGINEERING & TECHNOLOGY, GR. NOIDA	Session: 2025-26
		IV Semester
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING		

Database Management Systems Lab (BCSE0452Z)

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S.No.	Program Title	Date	Signature	Grade
1	Understand and implement the different ER diagram notation with their relationship and Cardinalities.			
2	Creating ER Diagram for company Database. Company database have entities like employee, departments, projects and dependents also implement the relationship and cardinalities between the entities with their relevant attribute.			
3	Implement DDL, DML, DCL & TCL commands.			
4	Implementation of I/O Constraint: Primary Key, composite primary key, Foreign Key with on delete set null and on delete set null constraint, Unique Key.			
5	Implementation of Business Constraint: Null, Not Null, Default, Check.			
6	Practicing Queries using Like, Between, Aliases, distinct Operator & Predicate. And Implement Aggregate Functions.			
7	Implementation of Queries using Where, Group by, Having and Order by Clause.			
8	Create a table EMPLOYEE with following schema:-(Emp_no, E_name, E_address, E_ph_no, Dept_no, Dept_name, Job_id, Designation, Salary) Write SQL statements for the following query. - List the E_no, E name, Salary of all employees working for MANAGER. - Display all the details of the employee whose salary is more than the Sal of any IT PROFF. - List the employees in the ascending order of Designations of those joined after 1981. - List the employees along with their Experience and Daily - List the employee who are either 'CLERK' or 'ANALYST'. - List the employees who joined on 1-MAY-81, 3-DEC-81, 17-DEC-81. - List the e_name those are starting with 'S'.			

	viii. Display total salary spent for each job category. ix. Display lowest paid employee details under each manager. x. Display number of employees working in each department and their department name. xi. Display the details of employees sorting the salary in increasing order. xii. Show the record of employee earning salary greater than 16000 in each department. xiii. Add constraints to check, while entering the empno value (i.e) empno > 100. xiv. Define the field DEPTNO as unique. Create a primary key constraint for the column (EMPNO).			
9	Implementation of Queries using set theory operators UNION, INTERSECT, MINUS.			
10	Implementation of Queries using Inner Join:- Natural Join , Equi Join & Non Equi Join, Outer Join			
11	Implementation of Queries nested Queries or Sub Queries: - IN, NOT IN, Exists, Not Exists, All and Any.			
12	1. Apply the set theory operators, join's and nested queries on company database (Case Study-1) Write the SQL Queries for the following statement. I. Retrieve the names of employees in department 5 who work more than 10 hours per week on the 'ProductX' project. II. List the names of employees who have a dependent with the same first name as themselves. III. Find the names of employees that are directly supervised by 'Franklin Wong'. IV. For each project, list the project name and the total hours per week (by all employees) spent on that project. V. Retrieve the names of all employees who work on every project controlled by department 5. VI. Retrieve the names of all employees who do not work on every project VII. For each department, retrieve the department name, and the average salary of employees working in that department. VIII. Retrieve the average salary of all female employees. IX. Find the names and addresses of all employees who work on at least one project located in Houston but whose department has no location in Houston. X. List the last names of department managers who have no dependents. Retrieve the names of all employees who work in the			

	department that has the employee with the highest salary among all employees.			
13	Understand and implement the Database Connectivity with Java/Python etc. programming language.			
14	Implementation and apply all the set theory operators, join and nested queries concept on Case study 1. - Make a list of all project members for projects that involve an employee whose name is SCOTT either as a worker or as a manager of the department that controls the project. - To retrieve the Social Security numbers of all employees who either work in department 5 or directly supervise an employee who works in department 5. - To retrieve the SSN of all employee who work as a supervisor not a manager. - We want a list of all employee names as well as the name of the departments they manage if they happen to manage a department; if they do not manage one, we can indicate it with a NULL value. - Retrieve the names of employees who have no dependents. - List the names of all employees with two or more dependents. - List the names of managers who have at least one dependent. - Retrieve the names of all employees who do not have supervisors. Retrieve the name of each employee who has a dependent with the same Last name as the employee.		-	
15	Implementation of Indexing, Views and sequence.			
16	- Write a PL/SQL Program to Add Two Numbers - Write PL/SQL Program for Fibonacci Series - Write PL/SQL Program to Find Greatest of Three Numbers.			
17	Write a PL/SQL code block to calculate the area of a circle for a value of radius varying from 3 to 7. Store the radius and the corresponding values of calculated area in an empty table named Areas, consisting of two columns Radius and Area.			
18	Write a PL/SQL code block that will accept an account number from the user, check if the users balance is less than the minimum balance, only then deduct Rs.100/- from the balance.			
19	Create a row level trigger for the customers table that would fire for INSERT or UPDATE or DELETE operations performed on the CUSTOMERS table. This trigger will display the salary difference between the old values and new values:			

20	Implementation of commit and rollback statement with amount transfer example.			
21	Implementation array, indexing, transaction concept on Case study 1. I. Implementation of Array Functions & Operators II. Implementation of Sequence • Creating Sequences • Modifying a Sequence Definition • Removing Sequences III. Implementation of Views • Creating Simple and Complex Views • Modifying Views • Removing Views IV. Implementation of Indexes • Manual and Automatic Indexes • Creating Indexes Removing Indexes			
22	Study of Open Source NOSQL Database and installation of MongoDB			
23	Implementation of the MongoDB Shell commands			
24	Implementation of the CRUD Operation in MongoDB			
25	Implementation of Aggregate in MongoDB			
26	Implementation of case Study on different domain I. E-commerce Platform II. Inventory Management III. Railway System IV. Hospital Data Management V. Voice-based Transport Enquiry System VI. SMS-based Remote Server Monitor system Banking System			